Congestion Games With Player-Specific Utility Functions and Its Application to NFV Networks

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***Abstract*— In this paper, a variation of the congestion situation is considered and a new game, the congestion game with player-specific (CGPS) utility functions, is proposed. This paper is motivated by some application scenarios that rule out the possibility of employing the existing game model to study such congestion situations. The CGPS game is characterized by adding a player-specific term and a weighted parameter with respect to the utility function. By using the semitensor product of matrices, the algebraic representation of the CGPS game is given and the existence of the weighted potential function is proved. Finally, the results are applied to solve the service chain composition problem in network function virtualization (NFV) and to analyze the effect of player-specific function on service chain configuration in NFV.**

***Note to Practitioners*—This paper is motivated by resource allocation problems in congestion networks where strategic users behave selfishly and aim at optimizing their own individual utility in the absence of a central controller. Compared with the centralized algorithms of poor reliability and scalability, game-theoretic control provides a promising distributed approach for resource allocation. In the game-theoretic framework, the existence and seeking of the desired solution are important issues. In this paper, a novel model is established to extend the utility functions space guaranteeing the existence of the solution. The developed utility design is used to capture users' different sensitivities to the effects of the network system. Simultaneously, it is more meaningful from the view of engineering to design the utility functions so that the desirable behavior is reachable. We also give an explicit scheme to seek the desired solution. The proposed model is finally applied to the service chain composition problem in NFV, of which the aim is to find the best service chain of users that accommodates their individual requirements. The proposed model shows reliable and effective.**

***Index Terms*— Algebraic formulation, congestion games, Nash equilibrium (NE), player-specific utility functions, semitensor product (STP) of matrices.**

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I. Introduction

CONGESTION situations originate from transportation engineering [1], where users behave selfishly and aim at optimizing their own individual welfare in the absence of a centralized traffic management. Rosenthal [2] first proposed the concept of congestion game to study the congestion situation rigorously. Rosenthal also proved that each congestion game possesses a Nash equilibrium (NE) by constructing a potential function. Based on Rosenthal's result, Monderer and Shapley [3] defined the class of potential games and also shown the equivalence between the potential game and congestion game.

The congestion game framework is well suited to model situations involved with resource competition [4]. Therefore, congestion games are used widely in practical applications. Marden *et al.* [5] used potential game to achieve cooperative control. An atomic congestion game [6] was introduced to model the traffic flow on a road over various time intervals of the day. These investigations are related to the NE [7], where none of the players can reduce its individual cost by a unilateral move. The computing complexity and the convergence speed of NE for congestion game were investigated in [8] and [9].

As pointed out in [10], a natural open question is to decide whether there are further utility functions guaranteeing the existence of an exact or weighted potential. Milchtaich and Shapley [11] proposed a weighted congestion model, where every player makes a different effect to the congestion on the chosen facilities. In contrast to unweighed congestion games, weighted congestion games do not always admit an NE. In [12], it was proved that there may be no NE even for a weighted congestion game with facilities costs being linear functions of their users. On the positive side, Panagopoulou and Spirakis [13] proved the existence of a weighted potential function for the case that all costs are determined by exponential functions. Facchini *et al.* [14] considered the congestion situation with incomplete information. The studies discussed above demand that all the players have the same sensitivity to the action. However, this assumption might not be realistic since players may have different sensitivities to the same action [14]. In this paper, we propose a new congestion game, the congestion game with player-specific (CGPS), to spread out players' behaviors by adding a weighted parameter and a player-specific term. The addition of these new components is based on the observation that, apart from the utility generated

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by the facilities, there is also an extra utility generated by the players themselves.

As an application of the proposed CGPS, consider, for example, the service chain composition problem in network function virtualization (NFV) [15]. NFV is a network architecture that visualizes entire classes of network node functions into software to create network services (NS). Each network user aims to minimize the cost for its flow that consists of network prices, communication latencies, and congestion costs. Although Ren and Meng [16] and Jocelyne *et al.* [16], [17] have proposed an optimization approach to design novel orchestration mechanisms to reduce the congestion and optimally control, the game theory provides a natural framework for seeking the best operational point of the system resulting from individual decisions [18]. For instance, the service chain composition problem of different bite rates has been investigated by leveraging game theory in [19]. However, the method presented in [19] is no longer applicable to the situation with different flow qualities. Taking account of the player-specific performance criteria, we solve the service chain composition problem with different flow qualities from the CGPS game prospective.

The main tool used in our paper is semitensor product (STP) of matrices [20], which is a generalization of conventional product to two arbitrary matrices. Applying STP to finite game theory is a very promising direction, and many excellent results have been obtained, such as the potential equation [21], [22], control of networked evolutionary games [23], and analysis of dynamic games with polytope strategy sets [24]. In addition, the algebraic state space approach based on STP provides a convenient platform for finite-valued dynamical systems [25]–[28]. It has been successfully applied to the analysis and control of Boolean networks [29]–[32] and mix-valued logical networks [33]–[36].

The main contribution of this paper is summarized as follows. First of all, we develop a new game by allowing a player-specific scaling of the private cost and adding a player-specific term to model congestion situation with player-specific cost. It should be noted that the role of the weight vector in our model is different from that in [11]. Second, we prove that the CGPS game admits pure strategy NE by inducing the integral of utility function [3] equal to zero. By leveraging on STP, the algebraic representation and potential function of the CGPS game are presented. Based on these results, we establish the existence of a potential function which depends only on the utility functions, and is independent of the action set, and the number of players. Finally, we apply the obtained results to the service chain composition problem in NFV. In view of the fact that the NFV composition problem is a special case of CGPS game with several layers, an algorithm is proposed to implement the calculation for the potential function. The numerical example and simulation verify the effectiveness of the algorithm.

The remainder of this paper is organized as follows. In Section II, the associated notations and preliminaries are provided. Section III presents the problem statement and motivates with an example the need for the CGPS game. Then, the algebraic structures and potential function of CGPS game are obtained by STP. In Section IV, the CGPS game is applied to investigate the service chain composition problem in NFV in which network users have player-specific cost functions. Section V is a concise conclusion.

II. Preliminaries

A. Notations

For statement ease, we introduce the notations which will be used in the sequel.

1) $\mathcal{R}^+$, $\otimes$, $\prod$, $\langle .,. \rangle$: the set of positive real number, Kronecker product, Cartesian product, and inner product.
2) $\mathcal{R}_{m\times n}$: the set of $m \times n$ real matrices.
3) $Col(A)$ (Row(A)): the set of columns (rows) of matrix A; $A(i,j)$: element in the ith row and jth column of matrix A.
4) $\delta_n^i := Col_i(I_n)$, and $\triangle_n := \{\delta_n^i | i = 1, 2, \ldots, n\}$.
5) $D_k := \{1, 2, \ldots, k\}$, and $D_k^n := \underbrace{D_k \times \cdots \times D_k}_{n}$.
6) A matrix $M \in \mathcal{R}_{m\times n}$ is called a logical matrix if the column of M is in the form of δ_n^k. By definition, it can be expressed as $M = [\delta_n^{i_1}, \delta_n^{i_2}, \ldots, \delta_n^{i_m}] \in \mathcal{R}_{n\times m}$. In this paper, it is briefly denoted as $M = \delta_n[i_1, i_2, \ldots, i_m]$.
7) $\mathbf{1}_n := \underbrace{(1, 1, \ldots, 1)}_{n}^T$, and $\mathbf{O}$ is zero matrix.
8) $\complement_U A$: the relative complement of set A in U.
9) $|v|$: the cardinality of v.

B. Semitensor Product of Matrices

Here, we give some necessary preliminaries on the STP of matrices. We refer to [20] for details.

Definition 1 [20]: Let $A \in \mathcal{R}_{m\times n}$, $B \in \mathcal{R}_{p\times q}$, and $t = lcm\{n, p\}$ be the least common multiple of n and p. The STP of A and B is defined as

$$A \ltimes B := (A \otimes I_{t/n})(B \otimes I_{t/p}) \in \mathcal{R}_{mt/n \times qt/p}. \tag{1}$$

When $n = p$, Definition 1 degenerates to the conventional matrix product AB. Thus, the symbol $\ltimes$ may be omitted without causing confusion. It is worth noting that STP keeps the main properties of conventional product.

Next, we introduce two dummy matrices, which can help us freely add dummy variables to a logical algebraic expression.

Proposition 1 [20]: Let $X \in \triangle_m$ and $Y \in \triangle_n$. Then

$$D_r[m,n]XY = X \tag{2}$$
$$D_f[m,n]XY = Y \tag{3}$$

where $D_r[m,n] := I_m \otimes \mathbf{1}_n^T$ and $D_f[m,n] := \mathbf{1}_m^T \otimes I_n$.

Assume $x = i \in D_k$. By identifying $i \sim \delta_k^i, i = 1, 2, \ldots, k$, we have $x = \delta_i^k \in \triangle_k$.

Definition 2 [21]: A function $f(x_1, x_2, \ldots, x_n) : D_k^n \mapsto \mathcal{R}$ is called a k-valued pseudo-logical function.

Theorem 1 [20]: Let $f(x_1, x_2, \ldots, x_n) : D_k^n \mapsto \mathcal{R}$ be a k-valued pseudo-logical function, where $x_j \in D_k$. Then there exists a unique row vector $V^f \in R_{k^n}$, called the structure vector of f, such that

$$f(x_1, x_2, \ldots, x_n) = V^f \ltimes_{j=1}^n x_j. \tag{4}$$

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C. Normal Form Games and Potential Games

Definition 3 [7]: A normal form game is a tuple $G = \{N, A, u\}$, where:

1) $N := \{1, 2, \dots, n\}$ is the set of players;
2) A_j is the action set of player j, $j = 1, \dots, n$, and $A = \prod_{j=1}^{n} A_j$ is called the action profile set of G;
3) $u = (u_1, \dots, u_n)$, $u_j : A \mapsto \mathcal{R}$ is the utility function of player j, $j = 1, \dots, n$.

We use $a_j \in A_j$ to represent an action of player j and $a = \prod_{j=1}^{n} a_j \in A$ represent an action profile in game G. We use u_j to represent player j's cost function in the sequel, then the action profile $a^* = (a_j^*, a_{-j}^*)$ is called the Nash equilibrium if for every player $j \in N$ and all $a_j \in A_j$

$$u_j\left(a_j^*, a_{-j}^*\right) \le u_j\left(a_j, a_{-j}^*\right)$$

where a_{-j} means the set of the actions by the players except for the player j.

Definition 4 [3]: Consider a normal form game $G = \{N, A, u\}$, G is called a weighted potential game if there exist a function $P : A \mapsto \mathcal{R}$ called potential function and a vector $\mathbf{w} = (w_1, w_2, \dots, w_n) \in \mathcal{R}^n$ such that

$$u_j(a_j, a_{-j}) - u_j(a_j', a_{-j}) = w_j(P(a_j, a_{-j}) - P(a_j', a_{-j}))$$

for every $j \in N$, $a_{-j} \in A_{-j}$ and $a_j, a_j' \in A_j$, and we call the game as $\mathbf{w}$-potential game.

Remark 1: Furthermore, game G is called a (exact) potential game when $\mathbf{w} = \mathbf{1}_n$.

Monderer and Shapley [3] characterized the existence of the exact potential game by the use of certain cycles defined below. Consider a normal form game $G = \{N, A, u\}$, an arbitrary action profile circle with a length of four $\phi : E \to B \to C \to D \to E$ where $E = (a_i, a_j, a_{-(i,j)})$, $B = (a_i', a_j, a_{-(i,j)})$, $C = (a_i', a_j', a_{-(i,j)})$, $D = (a_i, a_j', a_{-(i,j)})$, with $i, j \in N$, $a_i, a_i' \in A_i$, $a_j, a_j' \in A_j$, $a_{-(i,j)} \in A_{-(i,j)}$ is the action profile of all players except i and j. Let the integral of utility function along ϕ be defined as

$$\begin{aligned} I(\phi, u) = u_i(B) - u_i(E) + u_j(C) - u_j(B) \\ + u_i(D) - u_i(C) + u_j(E) - u_j(D) \end{aligned} \tag{5}$$

where $A_{-(i,j)}$ means the set of the actions by the players except for the players i and j.

Theorem 2 [3]: Let $G = \{N, A, u\}$ be a normal form game. Then, G is an exact potential game if and only if $I(\phi, u) = 0$ for an arbitrary action profile circle ϕ of length 4.

III. Congestion Games Review and Problem Formulation

A. Brief Review of Congestion Games

In this section, we provide a brief review of congestion games and weighted congestion games.

Definition 5 [2]: A congestion game is a tuple $\Gamma = \{N, M, (\Sigma_j)_{j \in N}, (c_l)_{l \in M}\}$, where:

1) $N := \{1, 2, \dots, n\}$ is the set of players;
2) $M := \{e_1, e_2, \dots, e_m\}$ is the set of facilities;

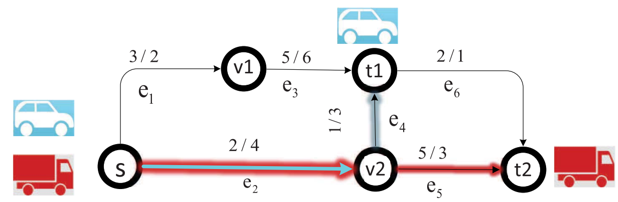

Fig. 1. Player-specific utilities congestion situation. The notation a/b on the road segment $e_l (l = 1, \dots, 6)$ means the car has maintenance cost a, and the truck has cost b.

3) for each player $j \in N$, his action set $\Sigma_j \subseteq 2^M$ is a finite family of subsets of M, consisting of the alternative routes from player j's starting point to its destination;
4) $c_l : N \to \mathcal{R}$ is the cost function of facility l and is assumed to be nondecreasing in general.

For any action profile $a = (a_1, \dots, a_n) \in \Sigma_1 \times \cdots \Sigma_n$, where a_i is identified with the set of facilities used by the player i:

1) $l \in a_j$ suggests that player j selects facility l under a;
2) $n_l(a) := |\{j \in N | e_l \in a_j\}|$ is the total number of players using facility e_l;
3) $c_l(n_l(a))$ is the cost of using facility e_l.

A player in this game aims to minimize its total cost: the total cost over all facilities that its action involves is given by

$$u_j(a) := \sum_{e_l \in a_j} c_l(n_l(a)). \tag{6}$$

In the model considered so far, all players are equally affected by the congestion. However, in some practical scenarios, this assumption may not be realistic. A scenario was discussed in [14], where the utilities of decision makers depend on their call cost per minute and specific rewards. In this kind of scene, the utilities, in general, have a weight parameter and a player-specific argument since different players experience different levels of influence and suffer different extra costs (rewards) even when using the same facility. It is easy to see that the above-mentioned congestion game cannot depict such cases in [14]. Note that under Definition 5, the utility functions are facility-dependent but player-independent. The next example justifies this claim in detail.

Example 1: Consider a transportation network shown in Fig. 1, where a sedan car and a heavy truck transit fresh products from its starting point to destination using road segments from segments sets $\{e_1, e_2, e_3, e_4, e_5, e_6\}$, respectively. The car and the truck have the same impact on traffic congestion, but the fuel consumption of the latter is twice that of the former in unit time. If we assume the unit fuel consumption of the car is "1," then the unit fuel consumption of the truck is "2." The fuel consumption costs of the vehicles increase with the number of users using the related road. The utility of the vehicle consists of its fuel consumption and maintenance cost of keeping products fresh.

Following Definition 5, we model this transportation system as a congestion game $\Gamma = \{N, M, (\Sigma_j)_{j \in N}, (c_l)_{l \in M}\}$. We consider segmented highways as facilities and use player 1 and player 2 to denote the car and truck, respectively.

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Then $N = \{1, 2\}$, $M = \{e_1, e_2, e_3, e_4, e_5, e_6\}$. Since the starting point of the car and truck is **s** and destinations are **t1** and **t2**, we have $\Sigma_1 = \{(e_1, e_3), (e_2, e_4)\}$ and $\Sigma_2 = \{(e_1, e_3, e_6), (e_2, e_4, e_6), (e_2, e_5)\}$. The cost function of roads is $c_l(n_l(a)), l = 1, \ldots, 6$. Given the car's route $a_1 = (e_1, e_3)$ and truck's route $a_2 = (e_2, e_5)$, We take the utility of player 2 (truck) to illustrate the calculation process. The car's maintenance costs on road e_1, e_3 are "3" and "5," respectively. The truck's maintenance costs on road e_2, e_5 are "4" and "3," respectively. The unit fuel consumption of the car and the truck is "1" and "2," respectively. We have the utilities of the car and the truck as follows:

$$u_1 = 3 + 5 + 1 * [c_1(n_1(a_1, a_2)) + c_3(n_3(a_1, a_2))]$$

$$u_2 = 4 + 3 + 2 * [c_2(n_2(a_1, a_2)) + c_5(n_5(a_1, a_2))].$$

Obviously, the utility function in (6) cannot describe the utilities of above-mentioned congestion situation, that is, Definition 5 cannot capture this kind of congestion situations. To depict and analyze this kind of congestion game, we would need to generalize Definition 5.

B. Problem Formulation and the Existence of Potential Function

We introduce the CGPS utility functions as a generalization. Before giving the formal definition of our generalized congestion game (CGPS), we first introduce the associated congestion form described by

$$F = \{N, M, (\Sigma_j)_{j\in N}, (r_j)_{j\in N}, (c_l)_{l\in M}, \mathbf{w}\}.$$

Terms $N, M, (\Sigma_j)_{j\in N}, (c_l)_{l\in M}$ in F are consistent with Definition 5. r_j is the additional player-specific function (e.g., a car in Example 1 bears the maintenance cost depending on its own route). $\mathbf{w} = (w_1, w_2, \ldots w_n) \in \mathcal{R}_{1\times n}$, and w_j is the specific scaling of player j's private cost and is strictly positive in general.

Then we can formally define the generalized congestion game.

Definition 6: A CGPS utility functions arising from the congestion form $F = \{N, M, (\Sigma_j)_{j\in N}, (r_j)_{j\in N}, (c_l)_{l\in M}, \mathbf{w}\}$ is a tuple $\Gamma = \{N, M, (\Sigma_j)_{j\in N}, u\}$, where:

1) $N := \{1, 2, \ldots, n\}$ is the set of players;
2) $M := \{e_1, e_2, \ldots, e_m\}$ is the set of facilities;
3) $\Sigma_j \subseteq 2^M$: action set of player j, and $a_j \in \Sigma_j$ is the action of player j;
4) $u = (u_1, \ldots, u_n)$, and $u_j : \Sigma \mapsto \mathcal{R}$ is the utility function of player j, which is the sum of his facilitiy cost and extra cost. For an action profile $a = (a_1, \ldots, a_n) \in \prod_{j=1}^n \Sigma_j$, player$j$'s utility is given by

$$u_j(a) := r_j(a_j) + w_j \sum_{e_l\in a_j} c_l(n_l(a)) \quad (7)$$

where $r_j(a_j)$ is the addition player-specific utility of player j under his action a_j, w_j is the weighted parameter of player j, and $n_l(a)$ denotes the number of players using facility e_l under action profile a.

Next, we show that every CGPS game with arbitrary player-specific utility function $(r_j)_{j\in N}$ and weight vector $\mathbf{w} \in \mathcal{R}_{1\times n}$ is a weighted potential game. To derive this result, we use a scaling technique that transforms a $\mathbf{w}$-potential game into an exact potential game. Here, we give the axiomatic characterization for the scaling technique, adopted in [37], on the verification of the weighted potential games.

Proposition 2: For a given vector $\mathbf{w} = (w_1, w_2, \ldots w_n) \in \mathcal{R}_{1\times n}$, a normal form game $G = \{N, A, u\}$ is a $\mathbf{w}$-potential game if and only if $I(\phi, u) = 0$ for all arbitrary action profile circle ϕ of length 4 in the game $\tilde{G} = \{N, A, \tilde{u}\}$ with utility functions $\tilde{u}_j = u_j/w_j (j \in N)$.

Proof: It is a obvious observation that the existence of a $\mathbf{w}$-potential game is equivalent to the existence of a vector $\mathbf{w}$ such that the game $\tilde{G} = \{N, A, \tilde{u}\}$ with utility functions $\tilde{u}_j = u_j/w_j (j \in N)$ is an exact potential game. Based on Definition 4 and Remark 1, we get the proposition. ■

In the following, we will use this characterization to verify that the CGPS games are weighted potential games.

Theorem 3: Let $\Gamma = \{N, M, (\Sigma_j)_{j\in N}, u\}$ be a CGPS game. Then Γ is a weighted potential game.

Proof: To certificate the CGPS games are weighted potential games, it is sufficient to deduce that for arbitrary given action profile circle ϕ of length 4, the integral of utility $I(\phi, u)$ in (5) equals 0 in game $\tilde{\Gamma} = \{N, M, (\Sigma_j)_{j\in N}, \tilde{u}\}$ with $\tilde{u}_j = u_j/w_j (j \in N)$. We fix $i, j \in N$, $a_i, a_i' \in \Sigma_i, a_j, a_j' \in \Sigma_j$ arbitrarily and use the notation $\sum_{X,Y}$ for $\sum_{f\in X\cup Y}(X, Y \subseteq \Sigma_j)$. For a facility $l \in M$, we define

$$\tilde{n}_l(a) := |\{k \in N \setminus \{i, j\} | l \in a_k\}|$$

as the number of players using the facility in the partial action profile $a_{-(i,j)}$.

Fixed an action profile circle $\phi : E \to B \to C \to D \to E$ in the game $\tilde{\Gamma} = \{N, M, (\Sigma_j)_{j\in N}, \tilde{u}\}$ with $\tilde{u}_j = u_j/w_j$, where $E = (a_i, a_j, a_{-(i,j)})$, $B = (a_i', a_j, a_{-(i,j)})$, $C = (a_i', a_j', a_{-(i,j)})$, $D = (a_i, a_j', a_{-(i,j)})$, $i, j \in N$, $a_i, a_i' \in \Sigma_i$, $a_j, a_j' \in \Sigma_j$, $a_{-(i,j)} \in \Sigma_{-(i,j)}$. From (6) and (7), we compute straightforwardly that

$$\begin{aligned} &I(\phi, \tilde{u}) \\ &= \tilde{u}_i(B) - \tilde{u}_i(E) + \tilde{u}_j(C) - \tilde{u}_j(B) \\ &\quad + \tilde{u}_i(D) - \tilde{u}_i(C) + \tilde{u}_j(E) - \tilde{u}_j(D) \\ &= \frac{r_i(a_i')}{w_i} + \sum_{l\in a_i'} c_l(n_l(B)) - \frac{r_i(a_i)}{w_i} - \sum_{l\in a_i} c_l(n_l(E)) \\ &\quad + \frac{r_j(a_j')}{w_j} + \sum_{l\in a_j'} c_l(n_l(C)) - \frac{r_j(a_j)}{w_j} - \sum_{l\in a_j} c_l(n_l(B)) \\ &\quad + \frac{r_i(a_i)}{w_i} + \sum_{l\in a_i} c_l(n_l(D)) - \frac{r_i(a_i')}{w_i} - \sum_{l\in a_i'} c_l(n_l(C)) \\ &\quad + \frac{r_j(a_j)}{w_j} + \sum_{l\in a_j'} c_l(n_l(E)) - \frac{r_j(a_j')}{w_j} - \sum_{l\in a_j} c_l(n_l(D)). \end{aligned} \quad (8)$$

For a_i, a_i', every facility $l \in M$ can be chosen by player i in both action a_i and action a_i' in one of these actions or not at all. The same holds for player j in both action a_j and

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	$a_j \setminus a_j'$	$a_j \cap a_j'$	$a_j' \setminus a_j$	$\complement_M(a_j \cup a_j')$
$a_i \setminus a_i'$	M_1	M_2	M_3	M_4
$a_i \cap a_i'$	M_5	M_6	M_7	M_8
$a_i' \setminus a_i$	M_9	M_{10}	M_{11}	M_{12}
$\complement_M(a_i \cup a_i')$	M_{13}	M_{14}	M_{15}	M_{16}

Fig. 2. Decomposition of M into 16 disjoint subsets.

action a_j'. We can thus decompose M into 16 disjoint sets $M_1, M_2, \ldots, M_{16}$. We take the first set to illustrate: M_1 comprises all facilities that are in $(a_i \setminus a_i') \cap (a_j \setminus a_j')$. The comprehensive description is given in Fig. 2.

In order to compute, for instance, the first two term of (8), we notice that, in action profiles E and B, the number of players using facility $l \in M_5 \cup M_6 \cup M_7 \cup M_8 \cup M_{13} \cup M_{14} \cup M_{15} \cup M_{16}$ are the same, i.e., $n_l(E) = n_l(B)$. Also, the numbers of players using facility $l \in M_1 \cup M_2$ equal $n_l(E) = \tilde{n}_l(a) + 2$, $n_l(B) = \tilde{n}_l(a) + 1$ in E, B, respectively. Similarly, for facility $l \in M_3 \cup M_4$, we have $n_l(E) = \tilde{n}_l(a) + 1$, $n_l(B) = \tilde{n}_l(a)$. For facility $l \in M_9 \cup M_{10}$, we have $n_l(E) = \tilde{n}_l(a) + 1$, $n_l(B) = \tilde{n}_l(a) + 2$. For facility $l \in M_{11} \cup M_{12}$, we have $n_l(E) = \tilde{n}_l(a)$, $n_l(B) = \tilde{n}_l(a) + 1$. These considerations lead to the following equation:

$$\begin{aligned} I(\phi, \tilde{u}) &= \sum_{l \in a_i'} c_l(n_l(B)) - \sum_{l \in a_i} c_l(n_l(E)) + \sum_{l \in a_j'} c_l(n_l(C)) \\ &\quad - \sum_{l \in a_j} c_l(n_l(B)) + \sum_{l \in a_i} c_l(n_l(D)) - \sum_{l \in a_i'} c_l(n_l(C)) \\ &\quad + \sum_{l \in a_j'} c_l(n_l(E)) - \sum_{l \in a_j} c_l(n_l(D) \\ &= \sum_{M_9, M_{10}} c_l(\tilde{n}_l(a) + 2) + \sum_{M_{11}, M_{12}} c_l(\tilde{n}_l(a) + 1) \\ &\quad - \sum_{M_1, M_2} c_l(\tilde{n}_l(a) + 2) - \sum_{M_3, M_4} c_l(\tilde{n}_l(a) + 1) \\ &\quad + \sum_{M_7, M_{11}} c_l(\tilde{n}_l(a) + 2) + \sum_{M_3, M_{15}} c_l(\tilde{n}_l(a) + 1) \\ &\quad - \sum_{M_5, M_9} c_l(\tilde{n}_l(a) + 2) - \sum_{M_1, M_{13}} c_l(\tilde{n}_l(a) + 1) \\ &\quad + \sum_{M_2, M_3} c_l(\tilde{n}_l(a) + 2) + \sum_{M_1, M_4} c_l(\tilde{n}_l(a) + 1) \\ &\quad - \sum_{M_{10}, M_{11}} c_l(\tilde{n}_l(a) + 2) - \sum_{M_9, M_{12}} c_l(\tilde{n}_l(a) + 1) \\ &\quad + \sum_{M_1, M_5} c_l(\tilde{n}_l(a) + 2) + \sum_{M_9, M_{13}} c_l(\tilde{n}_l(a) + 1) \\ &\quad - \sum_{M_3, M_7} c_l(\tilde{n}_l(a) + 2) - \sum_{M_{11}, M_{15}} c_l(\tilde{n}_l(a) + 1). \end{aligned}$$

Reordering the above-mentioned summation, many terms cancel out. Then we obtain $I(\phi, \tilde{u}) = 0$. ∎

C. Algebraic Representation of CGPS and Calculation of Potential Function

Without loss of generality, we assume the action set $A_j = 2^M = \Sigma_j \cup \complement_{2^M} \Sigma_j$, $j = 1, 2, \ldots, n$ and set $r_j(a_j) = \kappa$ when $a_j \in \complement_{2^M} \Sigma_j$, where κ is an arbitrary large number. Then the CGPS game can be rewritten as $\Gamma = \{N, M, (A_j)_{j \in N}, u\}$.

First, we consider the algebraic representation of the facility set $M = \{e_1, \ldots, e_m\}$. We can express M by the vector set Δ_m as $e_l \sim \delta_m^l$. For instance, in Example 1, given facility e_3, we have $e_3 \sim (0, 0, 1, 0, 0, 0)^T$ since $m = 6$.

Second, we consider the expression of players' actions. We introduce a new way to describe players' actions. We define

$$\mathbf{I}_l(a_j) := \begin{cases} 1, & e_l \in a_j \\ 0, & e_l \notin a_j. \end{cases} \tag{9}$$

Then the action of player j can be formally writen as

$$a_j = \sum_{l \in M} \mathbf{I}_l(a_j) \delta_m^l = [\mathbf{I}_1(a_j), \mathbf{I}_2(a_j), \ldots, \mathbf{I}_m(a_j)]^T. \tag{10}$$

In Example 1, we can rewritten $a_1 = (e_3, e_6)$ as $a_1 = (0, 0, 1, 0, 0, 1)$.

Remark 2: Under the algebraic representation in (10), the number of players using facility l under action profile a can be formulated by the following equation:

$$n_l(a) = \sum_{j=1}^{N} \langle a_j, \delta_m^l \rangle. \tag{11}$$

Next, we consider the algebraic representation of the player-specific utility function. We arrange the elements in $A_j = 2^M$, $j = 1, \ldots, n$ in lexicographic order

$$\begin{aligned} a_{j,t} &= [\mathbf{I}_1(a_{j,t}), \mathbf{I}_2(a_{j,t}), \ldots, \mathbf{I}_m(a_{j,t})]^T \text{ with} \\ t &= \sum_{l=1}^{m} 2^{(m-l)} \mathbf{I}_l(a_{j,t}) + 1. \end{aligned} \tag{12}$$

For each $j \in N$, we have

$$a_{j,1} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \quad a_{j,2} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{bmatrix}, \quad a_{j,3} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \\ 0, \end{bmatrix}, \ldots$$

$$a_{j,2^{m-1}} = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix}, \ldots, a_{j,2^{m-1}+1} = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \\ 0 \end{bmatrix}, \quad a_{j,2^m} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \\ 1 \end{bmatrix}.$$

Then player j's action set can be exactly expressed by the following matrix:

$$A_j = [a_{j,1}, a_{j,2}, \ldots, a_{j,2^m}] \in \mathcal{R}_{m \times 2^m}.$$

Remark 3: From (9) and (10), it is easy to know that the elements in $a_j : \mathbf{I}_l(a_j)$, $l = 1, \ldots, m$, are either 1 or 0. Furthermore, given the lexicographic order shown in (12), the index $t = (1, \ldots, 2^m)$ of a_j one to one corresponds to the index $[\mathbf{I}_1(a_j), \mathbf{I}_2(a_j), \ldots, \mathbf{I}_m(a_j)]$ as

$$\mathbf{I}_l(a_j) = \begin{cases} \left\lfloor \dfrac{t-1}{2^{m-1}} \right\rfloor, & l = 1 \\ \left\lfloor \dfrac{t - 1 - \sum_{k=1}^{l-1} 2^{m-k} \mathbf{I}_k(a_j)}{2^{m-l}} \right\rfloor, & l \in [2, m] \end{cases}$$

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where $\lfloor x \rfloor$ denotes the integer part of x.

Then the player-specific utility function $r_j(a_j)$, $j = 1, \ldots, n$ can be expressed into a vector as

$$R_j := (r_j(a_{j,1}), r_j(a_{j,2}), \ldots, r_j(a_{j,2^m})) \in \mathcal{R}_{1\times 2^m}. \quad (13)$$

Identify $0 \sim \delta_2^1$ and $1 \sim \delta_2^2$, then $\mathbf{I}_l(a_j) \in \Delta_2, a_j \in \Delta_{2^m}, j = 1, \ldots, n, a_j \in A_j$ in (10). By using this vector form of $\mathbf{I}_1(a_j)$ and a_j, it is easy to get the following fact.

Proposition 3: The player-specific utility function $r_j(a_j)$ in CPGS game can be expressed into a pseudo-logical function, as follows:

$$r_j(a_j) = R_j \ltimes a_j = R_j \ltimes_{i=1}^m \mathbf{I}_i(a_j)$$

where R_j is called the structure vector of player j's specific utility function $r_j(a_j)$.

Here, we use the calculation of R_1 in Example 1 to illustrate Proposition 3. From (10), we have action $a_1 = (e_1, e_3) = (1, 0, 1, 0, 0, 0)^T$. We further have its index $t = 1 * 2^5 + 1 * 2^3 + 1 = 41$ from (12). From Fig. 1, we can deduce $r_1(a_1) = 3 + 5 = 8$. By symmetry, we have $r_1(e_2, e_4) = r_1(a_{1,21}) = 3$. We conclude that

$$R_1(1,t) = \begin{cases} 3, & t = 21 \\ 8, & t = 41 \\ \kappa, & t \neq 21, 41. \end{cases}$$

Next, we consider the algebraic representation of the total facility cost. Before that, we give the matrix expression of the facility cost functions. We use $c_l(k)(l \in M, k \in N)$ to denote the cost of facility e_l when its users number equals k. Then the facility cost function c_l can be expressed as a vector

$$C_l := [c_l(1), c_l(2), \ldots, c_l(n)],\ l \in M. \quad (14)$$

Putting all facility cost functions together, we get the facility cost matrix C as

$$C := [C_1^T, C_2^T, \ldots, C_m^T]^T. \quad (15)$$

We arrange the elements $a \in A$ in lexicographic order

$$a^\tau = [a_{1,t_1}, a_{2,t_2}, \ldots, a_{n,t_n}] \quad (16)$$

with $\tau = \sum_{i=1}^n 2^{(n-i)m}(t_i - 1) + 1$.

From (14) and (15), we have the total facility cost of player $j \in N$ under action profile $a^\tau, \tau = 1, 2, \ldots, 2^{mn}$ as

$$c_F^j(a^\tau) = \sum_{e_l \in a_j} c_l(n_l(a^\tau)) = \sum_{e_l \in a_j} \mathrm{Row}_l C \delta_n^{n_l(a)}. \quad (17)$$

Then the total facility cost function can be expressed into a matrix as $C_F \in \mathcal{R}_{n\times 2^{mn}}$, where

$$C_F(j, \tau) := c_F^j(a^\tau).$$

Identify $a_{j,t} \sim \delta_{2^m}^t, t = 1, 2, \ldots, 2^m$, then $a^\tau \in \Delta_{2^{mn}}, \tau = 1, 2, \ldots, 2^{mn}$. Using this vector form of a_j and a^τ, it is easy to get the following proposition.

Proposition 4: The total facility cost of player j can be expressed into a pseudo-logical function, as follows:

$$c_F^j(a^\tau) = \mathrm{Row}_j(C_F) \ltimes_{i=1}^n a_i$$

where $\mathrm{Row}_j(C_F)$ is called the structure vector of player j's total facility cost.

Inspired of the original work of Cheng [21], the above-mentioned argument leads to the following proposition.

Proposition 5: Given a CGPS game $\Gamma = \{N, M, (A_j)_{j\in N}, u\}$, its reward matrix $R_\Gamma \in \mathcal{R}_{n\times 2^m}$ and cost matrix of players $C_F \in \mathcal{R}_{n\times 2^{mn}}$, then the structure vector V^{u_j}, $j \in N$ in game Γ can be obtained by

$$V^{u_j} = \begin{cases} R_j D_r[2^m, 2^{(n-1)m}] + \omega_j \mathrm{Row}_j(C_F), & j = 1 \\ R_j D_f[2^{(j-1)m}, 2^m] D_r[2^m, 2^{(n-j)m}] & \\ \qquad + \omega_j \mathrm{Row}_j(C_F), & j \neq 1. \end{cases} \quad (18)$$

Proof: Recalling Definition 6, we express (9) in its vector form as

$$V^{u_j} \ltimes_{i=1}^n a_i = R_j \ltimes a_j + \omega_j \mathrm{Row}_j(C_F) \ltimes_{i=1}^n a_i.$$

Given the dummy matrices' properties shown in proposition 1, we have

$$a_j = \begin{cases} D_r[2^m, 2^{(n-1)m}] \ltimes_{i=1}^n a_i, & j = 1 \\ D_f[2^{(j-1)m}, 2^m] D_r[2^m, 2^{(n-j)m}] \ltimes_{i=1}^n a_i, & j \neq 1. \end{cases}$$

Thus, we get the theorem. ■

Theorem 3 shows that the CGPS game must be a weighted potential game, that is, the CGPS game admits a weighted potential function. The following theorem gives the concrete form of the weighted potential function.

Theorem 4: Let $\Gamma = \{N, M, (\Sigma_j)_{j\in N}, u\}$ be a CGPS game. Then CGPS game admits a weighted potential function with weight vector $\mathbf{w} = (\omega_j)_{j\in N}$ given by

$$P(a) = \sum_{i\in N} \frac{R_i}{\omega_i} a_i + \sum_{l=1}^m \mathrm{Row}_l C \left[\mathbf{1}_{n(a)}^T \ltimes N(a)\right] \quad (19)$$

where $N(a) := ((\delta_n^1)^T, (\delta_n^2)^T, \ldots, (\delta_n^{n_l}(a))^T)^T$.

Proof: We fix $j \in N$, $a_j, a_j' \in A_j$ arbitrarily. We consider action profiles $a = (a_j, a_{-j})$, $a' = (a_j', a_{-j})$. We observe that the number of players using each facility $f \in M \setminus ((a_j \setminus a_j') \cup (a_j' \setminus a_j))$ is the same, i.e., $n_l(a) = n_l(a')$. On the other hand, for each facility $f \in a_j' \setminus a_j$, $n_l(a) = n_l(a') - 1$, while for each facility $f \in a_j \setminus a_j'$, $n_l(a') = n_l(a) - 1$.

Accordingly, from (17) and (18), we have

$$\begin{aligned} &u_j(a_j, a_{-j}) - u_j(a_j', a_{-j}) \\ &= V^{u_j} \ltimes a - V^{u_j} \ltimes a' \\ &= R_j(a_j - a_j') + \omega_j \left[\sum_{e_l \in a_j} \mathrm{Row}_l C \delta_n^{n_l(a)} - \sum_{e_l \in a_j'} \mathrm{Row}_l C \delta_n^{n_l(a')}\right] \\ &= R_j(a_j - a_j') + \omega_j \left[\sum_{e_l \in a_j \setminus a_j'} \mathrm{Row}_l C \delta_n^{n_l(a)} - \sum_{e_l \in a_j' \setminus a_j} \mathrm{Row}_l C \delta_n^{n_l(a')}\right]. \end{aligned} \quad (20)$$

In addition, (19) can be written as

$$P(a) = \sum_{i\in N} \frac{R_i}{\omega_i} a_i + \sum_{l=1}^m \sum_{k=1}^{n_l(a)} \mathrm{Row}_l C \delta_n^k. \quad (21)$$

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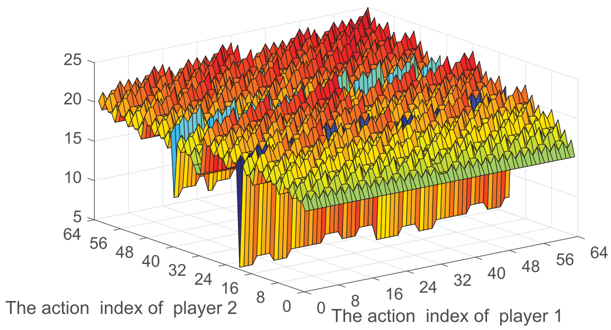

Fig. 3. Utility function u_1.

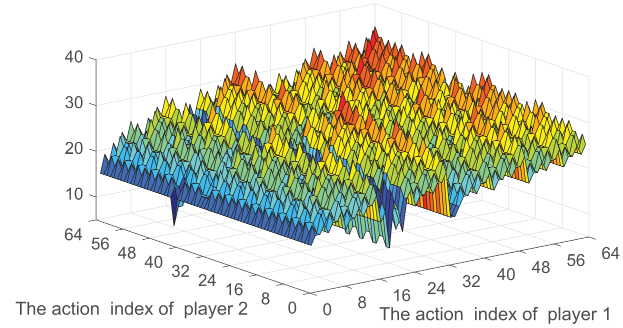

Fig. 4. Utility function u_2.

Using (21), we have

$$\begin{aligned}
&P(a) - P(a') \\
&= \frac{R_j}{w_j}(a_j - a_j') + \sum_{l=1}^{m}\sum_{k=1}^{n_l(a)} \mathrm{Row}_l C\delta_n^k \\
&\quad + \sum_{i\in N\setminus j}\frac{R_i}{w_i}(a_i - a_i') - \sum_{l=1}^{m}\sum_{k=1}^{n_l(a')} \mathrm{Row}_l C\delta_n^k \\
&= \frac{R_j}{w_j}a_j + \sum_{e_l\in a_j\setminus a_j'}\left(\sum_{k=1}^{n_l(a)} \mathrm{Row}_l C\delta_n^k - \sum_{k=1}^{n_l(a)-1} \mathrm{Row}_l C\delta_n^k\right) \\
&\quad - \frac{R_j}{w_j}a_j' + \sum_{e_l\in a_j'\setminus a_j}\left(\sum_{k=1}^{n_l(a')-1} \mathrm{Row}_l C\delta_n^k - \sum_{k=1}^{n_l(a')} \mathrm{Row}_l C\delta_n^k\right) \\
&= \frac{R_j}{w_j}(a_j - a_j') \\
&\quad + \left(\sum_{e_l\in a_j\setminus a_j'} \mathrm{Row}_l C\delta_n^{n_l(a)} - \sum_{e_l\in a_j'\setminus a_j} \mathrm{Row}_l C\delta_n^{n_l(a')}\right).
\end{aligned} \tag{22}$$

Finally, from (20) and (22), we obtain that

$$u_i(a_i, a_{-i}) - u_i(a_i', a_{-i}) = w_i(P(a_i, a_{-i}) - P(a_i', a_{-i})).$$

According to Definition 4, the function P is a weighted potential function for game Γ. ■

Remark 4: Based on Propositions 3 and 4, we can rewritten the potential function given by (19) as

$$P(a) = \sum_{j\in N}\frac{r_j(a_j)}{w_j} + \sum_{l\in M}\sum_{k=1}^{n_l(a)} c_l(k).$$

Remark 5: In fact, the form of the potential function in (19) includes that of the congestion game in Definition 5, that is, when the weighted vector $\mathbf{w} = \mathbf{1}_n$ and player-specific terms $r_j(a_j) = 0$ $(j = 1, \ldots, n)$, the CGPS game degenerates into the congestion game in Definition 5 and its potential function can be obtained by (19).

Based on the above-mentioned discussions, the calculation process of potential function of a CGPS game $\Gamma = \{N, M, (A_j)_{j\in N}, u\}$ can be summarized as follows. Given a CGPS game Γ, for each element $a^\tau (\tau = 1, 2, \ldots, 2^{mn})$, update the number of players using facility $l, l = 1, 2, \ldots, m$ by Remark 2, the facilities costs by (14) and the player-specific terms by (13). Finally, update the potential function value by (19).

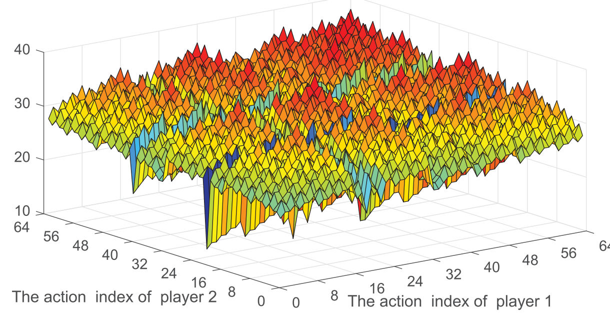

Fig. 5. Value of the potential function P_w.

Example 2: Recalling Example 1, we assume the cost of fuel consumption equals the number of users using the related road, i.e., $c_l(n_l(a)) = n_l(a)$ and the player-specific utility value $r_j(a_j) = \kappa = 15$ when $a_j \in \complement_{2^M}\Sigma_j$.

According to Definition 6, we model this transportation system into a congestion game $\Gamma = \{N, M, A, u\}$, where $N = \{1, 2\}$, $M = \{e_1, e_2, e_3, e_4, e_5, e_6\}$, $A_1 = A_2 = 2^M$. The fuel consumption cost function of roads is $c_l(n_l(a)) = n_l(a), l = 1, \ldots, 6$. From (18), (19), and (21), we have the value of utility functions u_1, u_2 and potential function P^w shown in Figs. 3–5, respectively.

From Fig. 5, we can find $\min P^w(a_1, a_2) = P^w(a_{1,21}, a_{2,19}) = 11.5$. Following Remark 3, we obtain the pure NE $a_1^* = (0, 1, 0, 1, 0, 0)^T$, $a_2^* = (0, 1, 0, 0, 1, 0)^T$ which corresponds to the path $\{(e_2, e_4), (e_2, e_5)\}$.

We consider the behaviors of the car and the truck to investigate the impact of the player-specific utility $r_j(a_j)$ on the congestion system. Although it will cause congestion, the car and the truck choose the same segmented road e_2 at NE point $\{(e_2, e_4), (e_2, e_5)\}$. As $w_1 = 1$, $w_2 = 2$, $c_l(n_l(a)) = n_l(a)$, and $r_j(a_j)$ is much larger than $c_l(n_l(a))$, the contribution generated by the term $w_j(c_l(n_l(a))$ is less relevant than $r_j(a_j)$, that is, high values of $r_j(a_j)$ and small values of w_j make the impact of facility congestion costs less important compared to that of player-specific utility.

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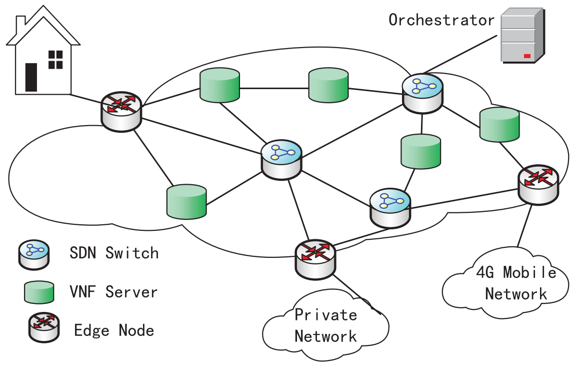

Fig. 6. Network scenario adopted from [19].

IV. Game-Theoretic Model of the Network Functions Virtualization

A. Model Description

In the past years, the two networking paradigms, software-defined networks [38] and network functions virtualization (NFV) [39], have been introduced to transform the ossified Internet in a more agile, flexible, and dynamic ecosystems. For this reason, NFV has recently gained momentum among network operators. See [15] for a survey of results and the state of the art on network function virtualization. Network functions virtualization is a network architecture that virtualizes entire classes of network node functions out of dedicated hardware devices and into software to create NS. Due to the gains in energy and space granted by virtualization, NFV is incrementally deployed by Internet service providers in their carrier networks.

As shown in Fig. 6, in this paper, we consider a network owned by a network orchestrator whose objective is to provide its customers with NS following the NFV approach. The main roles in the system are played by the network users, the virtual network function (VNF) servers and the network orchestrator [39]. Users generate traffic flows. NS with the associated levels of quality can be requested by flows. VNF servers are NFV-compliant nodes that provide VNF as a service in order to obtain some economic benefits. A VNF server can provide more than one service and decides the service price autonomously. The network orchestrator is responsible for management and orchestration of the whole system. More specifically, the network orchestrator makes the decision about the number of network users which have to run on the network to compose an NS, and that of the VNF servers which execute these instances, and their concatenation to achieve the service chains for each traffic flow in the network.

As pointed in [18], the content providers can provide video flows with different levels of bit rate and quality based on NFV telco operator platform. The problem of different bite rates has been considered by using game theory in [19]. In this paper, we leverage game theory to achieve the service chain composition problem in the scenario where flows have different levels of quality.

As discussed in [19], an NS can be modeled as an NFV service chain. Then we follow the framework in [19] and introduce some notations needed to describe the system. We characterize service chain by $\mathbf{F} = \{1, 2, \dots, F\}$. The VNF is executed on a VNF servers set $\mathbf{V} = \{1, 2, \dots, V\}$. For each server $\upsilon \in \mathbf{V}$, let $\mathbf{F}_\upsilon \in F$ be the subset of VNFs provided by the server υ. Without loss of generality, we will assume $\mathbf{F}_\upsilon = F, \forall \upsilon \in \mathbf{V}$.

Let $N = \{1, 2, \dots, n\}$ be the set of the users. Users can be either a user terminal or a local area network (LNA) (e.g., the home LAN in a residential environment) and have a different data synchronization rate w_i. Define the pair (s_i, t_i) to denote user i's position in the network, where s_i and t_i are the ingress and the egress nodes of user i, respectively. Each user is characterized by its data synchronization rate w_i and position (s_i, t_i).

Let $\mathbf{Z}$ be the set of all possible configurations for the NS. Thus, the cardinality of $\mathbf{Z}$ is $|\mathbf{Z}| = V^F$. Furthermore, let $\mathbf{z}_i \in \mathbf{Z}$ be the NS configuration chosen by the user $i \in N$. Specifically, $\mathbf{z}_i$ is defined as the F-tuple $\mathbf{z}_i = (\mathbf{z}_i(1), \mathbf{z}_i(2), \dots, \mathbf{z}_i(F))$, where $\mathbf{z}_i(f) \in \mathbf{V}$ is the VNF server, which has been chosen by user i to execute the function $f \in F$.

To take network prices into account, let $\mathbf{p}^{f-\upsilon} = (p_{f,\upsilon})_{f\in\mathbf{F},\upsilon\in\mathbf{V}}$ be the price matrix containing the prices between all VNF servers involved in the network service $\mathbf{F}$. For a given NS configuration $\mathbf{z}_i$ chosen by user i, the corresponding price $c_i^P(\mathbf{z}_i)$ can be written as

$$c_i^P(\mathbf{z}_i) = \sum_{f=1}^{F} p_{f,\mathbf{z}_i(f)} \tag{23}$$

where $p_{f,\mathbf{z}_i(f)}$ is the VNF price to execute an instance of the VNF f on the server $\mathbf{z}_i(f)$.

To consider the communication latencies between users and VNF servers, we define the following latency matrices. Let $\mathbf{D}^{\text{in}} = (d^{\text{in}}_{s_i,\upsilon})_{i\in N,\upsilon\in\mathbf{V}}$ be the ingress-to-server delay matrix, which contains the latencies between ingress nodes and the first VNF server in the network. Let $\mathbf{D}^{\upsilon-\upsilon} = (d_{\mathbf{z}_i(f-1),\mathbf{z}_i(f)})_{i\in N, f\in\mathbf{F}}$ be the interserver latency matrix between servers $\mathbf{z}_i(f-1)$ and $\mathbf{z}_i(f)$. Similarly, let $\mathbf{D}^{\text{out}} = (d^{\text{out}}_{\upsilon,t_i})_{i\in N,\upsilon\in\mathbf{V}}$ be the server-to-egress delay matrix which contains the delays between the last VNF server and egress nodes. Then the overall latency experienced by the player i when choosing the NS configuration $\mathbf{z}_i$ can be expressed as

$$c_i^D(\mathbf{z}_i) = d_{i,\mathbf{z}_i(1)} + \sum_{f=2}^{F} d_{\mathbf{z}_i(f-1),\mathbf{z}_i(f)} + d_{\mathbf{z}_i(F),i} \tag{24}$$

where the term $d_{i,\mathbf{z}_i(1)\in\mathbf{D}^{\text{in}}}$ is the delay between the ingress node s_i and the server $\mathbf{z}_i(1) \in V$, and the term $d_{\mathbf{z}_i(F),i} \in \mathbf{D}^{\text{out}}$ is the delay between the server $\mathbf{z}_i(F)$ and the egress node t_i.

The congestion cost $c_{\mathbf{z}_i(f)}$ experienced by the flow corresponding to the user i for function f on the chosen server $\mathbf{z}_i(f)$ depends on the congestion level $n_{\mathbf{z}_i(f)}$. We define the set of players in N, which have chosen the server υ to receive the function $f \in \mathbf{F}$ as

$$\Gamma_\upsilon^f(\mathbf{z}_i, \mathbf{z}_{-i}) := \{j \in N : \mathbf{z}_j(f) = \upsilon, \mathbf{z}_j(f) \in \mathbf{z}_j\}.$$

Then we have

$$n_{\mathbf{z}_i(f)}(\mathbf{z}_i, \mathbf{z}_{-i}) = \left|\Gamma_\upsilon^f(\mathbf{z}_i, \mathbf{z}_{-i})\right|. \tag{25}$$

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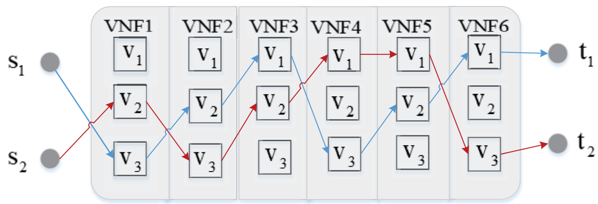

Fig. 7. Illustrative example of a possible service chain configuration.

It should be noticed that the congestion level is different from [19]. Since this paper considers the quality of the flows rather than the bit rate, the flows have the same impact on the congestion.

We have that the unit overall congestion costs experienced by the flow of user $i \in N$

$$c_i^C(\mathbf{z}_i, \mathbf{z}_{-i}) = \sum_{f=1}^{F} c_{\mathbf{z}_i(f)}(n_{\mathbf{z}_i(f)}(\mathbf{z}_i, \mathbf{z}_{-i})). \tag{26}$$

Considering different data synchronization rates of each player $w_i (i = 1, \ldots n)$, the total cost experienced by the flow represented by the user $i \in N$ can be expressed by the following cost function:

$$C_i(\mathbf{z}_i, \mathbf{z}_{-i}) = w_i c_i^C(\mathbf{z}_i, \mathbf{z}_{-i}) + c_i^P(\mathbf{z}_i) + c_i^D(\mathbf{z}_i). \tag{27}$$

For the sake of illustration, a possible service chain model is shown in Fig. 7. We present the system configuration where $(F = 6)$ VNFs compose the NS, $(V = 3)$ VNF servers are in the network, and two players choose the NS configuration for their flows.

B. Game Formulation of the NFV

If we regard the network users as players, we can model the NFV paradigm into a congestion game $\Gamma = \{N, M, (\Sigma_j)_{j \in N}, u\}$ in Definition 6 where N is the set of players, the facilities set M is the VNF servers set $\mathbf{V}$, and the action space of jth player Σ_j consists of all the possible NS configurations $\mathbf{Z}$ that can be chosen by the player $j \in N$. The utility function of player j defined in (7) corresponds to (27), where $c_i^D(\mathbf{z}_i) + c_i^P(\mathbf{z}_i)$ corresponds to the player-specific term $r_j(a_j)$, $w_i c_i^C(\mathbf{z}_i, \mathbf{z}_{-i})$ corresponds to the weighted term $w_j \sum_{l \in a_i} c_l(n_l(a))$, and the weights w_i are different data synchronization rates of the relative flows. A strategy profile for the game Γ consists of a tuple $\mathbf{z} = (\mathbf{z}_i, \mathbf{z}_{-i}) \in \mathbf{Z}^n$, where the cardinality of $\mathbf{Z}^n$ is $|\mathbf{Z}^n| = V^{nF}$.

From the above-mentioned discussion, we can see that the resulted NFV game is a special case of the CGPS game, where each route consists of F layers, and we give the weighted potential function of the resulted NFV game in Theorem 5. For the general CGPS game, we show its weighted potential function in Section III-C. However, Theorem 5 goes further than Remark 4 in Section III-C since each route in the resulted NFV game has F layers.

Theorem 5: The resulted NFV game $\Gamma = \{N, \mathbf{V}, \mathbf{Z}, u\}$ admits a weighted potential function with weight vector $\mathbf{w} = (w_j)_{j \in N}$ given by

$$P(\mathbf{z}) = \sum_{j \in N} \frac{c_j^D(\mathbf{z}_j) + c_j^P(\mathbf{z}_j)}{w_j} + \sum_{f=1}^{\mathbf{F}} \sum_{v \in \mathbf{V}} \sum_{k=1}^{n_v(\mathbf{z})} c_v(k). \tag{28}$$

Proof: For any given player $i \in N$, we consider action profile $\mathbf{z} = (\mathbf{z}_i, \mathbf{z}_{-i}), \mathbf{z}' = (\mathbf{z}_i', \mathbf{z}_{-i})$. Let $\mathbf{F}_{\mathbf{z}_i \mathbf{z}_i'} = \{f \in \mathbf{F} : \mathbf{z}_i(f) \neq \mathbf{z}_i'(f)\}$ be the set of functions which are executed on different servers when NS configurations $\mathbf{z}_i$ and $\mathbf{z}_i'$ are chosen by the player i.

Accordingly, from (27), we have

$$\begin{aligned} &C_i(\mathbf{z}_i, \mathbf{z}_{-i}) - C_i(\mathbf{z}_i', \mathbf{z}_{-i}) \\ &\quad = w_i c_i^C(\mathbf{z}_i, \mathbf{z}_{-i}) - w_i c_i^C(\mathbf{z}_i', \mathbf{z}_{-i}) \\ &\qquad + c_i^P(\mathbf{z}_i) + c_i^D(\mathbf{z}_i) - c_i^P(\mathbf{z}_i') - c_i^D(\mathbf{z}_i') \\ &\quad = w_i \sum_{f \in \mathbf{F}_{\mathbf{z}_i \mathbf{z}_i'}} (c_{\mathbf{z}_i(f)}(n_{\mathbf{z}_i(f)}(\mathbf{z})) - c_{\mathbf{z}_i'(f)}(n_{\mathbf{z}_i'(f)}(\mathbf{z}'))) \\ &\qquad + c_i^P(\mathbf{z}_i) + c_i^D(\mathbf{z}_i) - c_i^P(\mathbf{z}_i') - c_i^D(\mathbf{z}_i'). \end{aligned}$$

On the other hand, from (28), we have

$$\begin{aligned} &P(\mathbf{z}) - P(\mathbf{z}') \\ &\quad = \frac{c_i^P(\mathbf{z}_i) + c_i^D(\mathbf{z}_i)}{w_i} - \frac{c_i^P(\mathbf{z}_i') + c_i^D(\mathbf{z}_i')}{w_i} \\ &\qquad + \sum_{f \in \mathbf{F}_{\mathbf{z}_i \mathbf{z}_i'}} \left(\sum_{k=1}^{n_{\mathbf{z}_i(f)}} c_{\mathbf{z}_i(f)}(k) - \sum_{k=1}^{n_{\mathbf{z}_i'(f)}} c_{\mathbf{z}_i'(f)}(k) \right) \\ &\quad = \frac{c_i^P(\mathbf{z}_i) + c_i^D(\mathbf{z}_i)}{w_i} - \frac{c_i^P(\mathbf{z}_i') + c_i^D(\mathbf{z}_i')}{w_i} \\ &\qquad + \sum_{f \in \mathbf{F}_{\mathbf{z}_i \mathbf{z}_i'}} (c_{\mathbf{z}_i(f)}(n_{\mathbf{z}_i(f)}(\mathbf{z})) - c_{\mathbf{z}_i'(f)}(n_{\mathbf{z}_i'(f)}(\mathbf{z}'))). \end{aligned}$$

Finally, we obtain that

$$C_i(\mathbf{z}_i, \mathbf{z}_{-i}) - C_i(\mathbf{z}_i', \mathbf{z}_{-i}) = w_i(P(\mathbf{z}_i, \mathbf{z}_{-i}) - P(\mathbf{z}_i', \mathbf{z}_{-i}))$$

i.e., the function P is a weighted potential function for the resulted NFV game Γ. ■

Similar to the work in Section III-C, we arrange all possible NS configurations and strategies in lexicographic order. Thus, we can compute all the values of the potential function in sequence, and get the configuration corresponding to the maximum of the potential function easily.

Proposition 6: Assume the resulted NFV game $\Gamma = \{N, \mathbf{V}, \mathbf{Z}, u\}$. Then the elements in the action (NS configurations) set $\mathbf{Z}_i$ can be arranged as

$$\mathbf{z}_{i,t} = (\mathbf{z}_i(1), \mathbf{z}_i(2), \ldots, \mathbf{z}_i(F))$$

with $t = \sum_{f=1}^{F} V^{(F-f)}(\mathbf{z}_i(f) - 1) + 1, \mathbf{z}_i(f) = 1, 2, \ldots, V$. That is,

$$\begin{gathered} \mathbf{z}_1 = (v_1, v_1, \ldots, v_1), \mathbf{z}_2 = (v_1, v_1, \ldots, v_2), \ldots \\ \mathbf{z}_{V^F - 1} = (v_V, v_V, \ldots, v_{V-1}), \mathbf{z}_{V^F} = (v_V, v_V, \ldots, v_V). \end{gathered}$$

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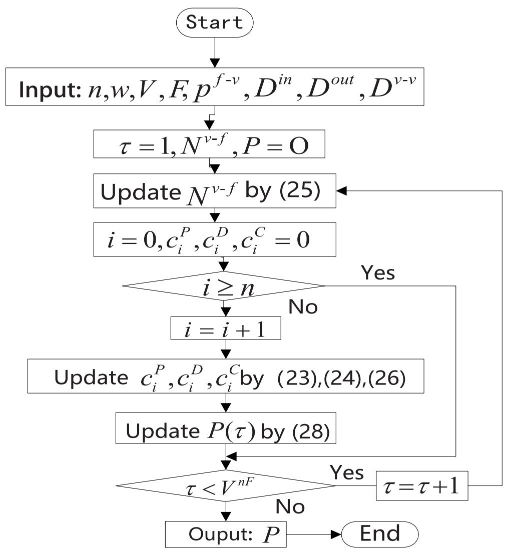

Fig. 8. Flowchart of the calculation for the potential function of the resulted NFV game.

Remark 6: The index $t = (1, \ldots, V^F)$ of $\mathbf{z}_{i,t}$ corresponds to the index $(t_1, t_2, \ldots, t_F)$ as

$$\mathbf{z}_i(f) = \left\lfloor \frac{t-1}{V^{(F-1)}} \right\rfloor + 1, \; f = 1$$

$$\mathbf{z}_i(f) = \left\lfloor \frac{t - 1 - \sum_{k=1}^{f-1} V^{(F-k)}(\tau_k - 1)}{V^{(F-f)}} \right\rfloor + 1, \quad \forall f = 2, \ldots, F$$

where $\lfloor x \rfloor$ denotes the integer part of x.

Similarly, we arrange the strategy profile $\mathbf{z} \in \mathbf{Z}^n$ in lexicographic order

$$\mathbf{z}^\tau = [\mathbf{z}_{1,t_1}, \mathbf{z}_{2,t_2}, \ldots, \mathbf{z}_{n,t_n}]$$

with $\tau = \sum_{i=1}^n V^{(n-i)F}(t_i - 1) + 1, t_i = 1, 2, \ldots, V^F$.

We define the matrix $\mathbf{N}^{\upsilon - f} \in \mathcal{R}_{V \times F}, \mathbf{P} \in \mathcal{R}_{1 \times V^{nF}}$, where $n_{\upsilon,f}$ denotes the number of players using the NFV server υ in the fth chain and $\mathbf{P}(\tau)$ denotes the potential function value of $\mathbf{z}^\tau$. For every player $i \in N$ and each element in the action profile set, we update the matrix $\mathbf{N}^{\upsilon - f}$, c_i^P, c_i^D, and c_i^C by (25), (23), (24), and (26) and get the potential function $\mathbf{P}$ by (28).

A flowchart of the practical calculation for the potential function of the resulted NFV game is summarized in Fig. 8.

C. Numerical Example

Example 3: To illustrate the problem model, we use the case based on the designed model in Fig. 7. The two users are a user terminal and a home LAN in a residential environment, and data synchronization rates are $w_1 = 2, w_2 = 1$. The price matrix and the interserver latency matrix are given as follows:

$$\mathbf{p}^{f-\upsilon} = \begin{bmatrix} 20 & 5 & 7 \\ 9 & 6 & 3 \\ 9 & 8 & 10 \\ 11 & 15 & 13 \\ 5 & 7 & 9 \\ 15 & 19 & 10 \end{bmatrix}, \quad \mathbf{D}^{\upsilon-\upsilon} = \begin{bmatrix} 5 & 7 & 3 \\ 6 & 9 & 8 \\ 10 & 5 & 12 \end{bmatrix}.$$

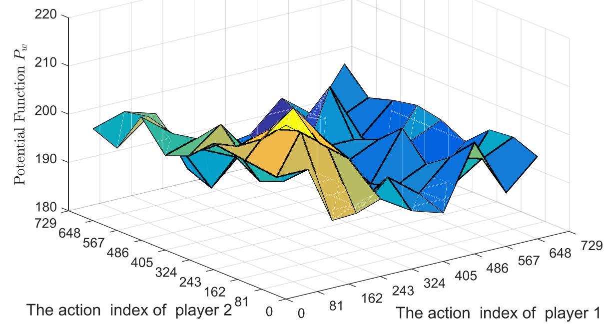

Fig. 9. Value of the potential function P_w.

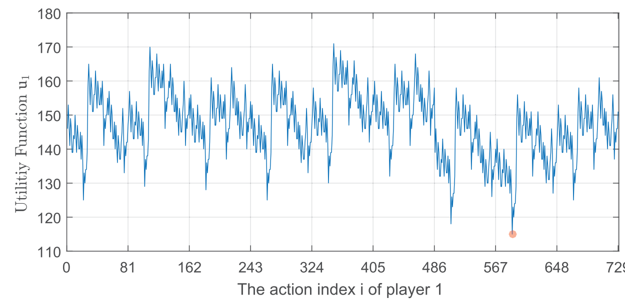

Fig. 10. Value of the utility function $u_1(\mathbf{z}_1, \mathbf{z}_2^*)$, when fixed $\mathbf{z}_2 = \mathbf{z}_2^* = (\upsilon_2, \upsilon_3, \upsilon_2, \upsilon_1, \upsilon_1, \upsilon_1)$.

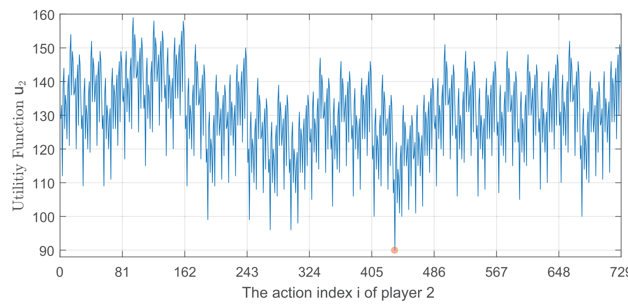

Fig. 11. Value of the payoff function $u_2(\mathbf{z}_1^*, \mathbf{z}_2)$, when fixed $\mathbf{z}_1 = \mathbf{z}_1^* = (\upsilon_3, \upsilon_2, \upsilon_1, \upsilon_3, \upsilon_2, \upsilon_3)$.

The server-to-egress delay matrix and the ingress-to-server delay matrix are

$$\mathbf{D}^{\text{out}} = \begin{bmatrix} 5 & 9 & 6 \\ 9 & 8 & 11 \end{bmatrix}, \quad \mathbf{D}^{\text{in}} = \begin{bmatrix} 7 & 4 & 9 \\ 6 & 9 & 3 \end{bmatrix}.$$

Considering an NS configuration $\mathbf{z}$, the congestion cost function is given as follows: $c_{\upsilon 1}(n_{\upsilon 1}(\mathbf{z})) = (n_{\upsilon 1}(\mathbf{z}))^3$, $c_{\upsilon 2}(n_{\upsilon 2}(\mathbf{z})) = 3n_{\upsilon 2}(\mathbf{z})$, and $c_{\upsilon 3}(n_{\upsilon 3}(\mathbf{z})) = (n_{\upsilon 3}(\mathbf{z}))^2$. Since $V = 3, F = 6$, the cardinality of $\mathbf{Z}$ is $|\mathbf{Z}^n| = V^F = 729$. Following the flowchart shown in Fig. 8, we get the potential function of the considered NFV model in Fig. 9.

From Fig. 9, we can find $\min P^w(\mathbf{z}_1, \mathbf{z}_2) = P^w(\mathbf{z}_{1,589}, \mathbf{z}_{2,435})$. Following Remark 6, we obtain the pure NE $\mathbf{z}_1^* = (\upsilon_3, \upsilon_2, \upsilon_1, \upsilon_3, \upsilon_2, \upsilon_1)$, $\mathbf{z}_2^* = (\upsilon_2, \upsilon_3, \upsilon_2, \upsilon_1, \upsilon_1, \upsilon_3)$. The service chain routes corresponding to $(\mathbf{z}_1^*, \mathbf{z}_2^*)$ are marked in Fig. 7, which are represented by blue and red lines, respectively. The NE allows network users to achieve their own service chain configurations according to their requirements.

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From the given price matrix ($\mathbf{p}^{f-v}$), the latency matrix ($\mathbf{D}^{v-v}$, $\mathbf{D}^{\text{out}}$, $\mathbf{D}^{\text{in}}$) and the congestion cost functions ($c_v(n_v(\mathbf{z}))$), we can see that the contribution generated by the price and latency is smaller than the congestion term. Thus, the network users select different servers in each layer to minimize congestion and improve their utilities.

Moreover, the strategy profile $(\mathbf{z}_1^*, \mathbf{z}_2^*)$ satisfies

$$115 = u_1(\mathbf{z}_1^*, \mathbf{z}_2^*) \le u_1(\mathbf{z}_1, \mathbf{z}_2^*), \quad \forall \mathbf{z}_1 \in A_1$$

$$90 = u_2(\mathbf{z}_1^*, \mathbf{z}_2^*) \le u_2(\mathbf{z}_1^*, \mathbf{z}_2), \quad \forall \mathbf{z}_2 \in A_2$$

which is shown in Figs. 10 and 11. From the perspective of the definition of NE, we verify the correctness of the result.

V. Conclusion

This paper considers the congestion situation with player-specific utility functions and the associated weighted congestion games, CGPS game. By resorting to STP of matrices, we obtain the algebraic representation and structure of CGPS game. We also prove that the CGPS game is a weighted potential game and give its potential function. Furthermore, we formulate the service chain composition problem in NFV networks with player-specific cost functions as a special case of CGPS game and obtain the weighted potential function of resulted NFV game. Moreover, an algorithm is provided to search the NE of the resulted NFV game. It should be pointed out that the complexity of the algorithms proposed in this paper needs further improvement, because the current result is just an direct process from the theoretical analysis.

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